MCAST

ASSESSMENT AND INTERNAL VERIFICATION FRONT SHEET (Individual Criteria)

(Note: This version is to be used for an assignment brief issued to students via Classter)

Course Title	**Bachelor of Arts (Honours) in Interactive Digital Media**		**Lecturer Name & Surname**	**Matthew Cumbo**	
Unit Number & Title	**CAIDM-506-2213 | API Systems**				
Assignment Number, Title / Type	**REST and Relax**				
Date Set	4th March 2025	**Deadline Date**	**29th May 2025**		
Student Name		**ID Number**		**Class / Group**	**INT 5.2**

Assessment Criteria	Maximum Mark
KU1 Describe the main differences between API types and architectural constraints for RESTful APIs.	10
KU2 Discuss and explain OAuth 2.0 in relation to APIs.	10
KU3 Identify OWASP API Security Top 10 and ways to mitigate against them.	10
KU4 Describe the function of mock servers when testing APIs.	10
AA1 Demonstrate best practices when designing APIs, specifically architectural and schema design.	10
AA2 Use cURL and Postman to make calls to an API.	10
AA3 Produce cohesive documentation outlining the specifications and use of a custom API.	10
AA4 Use Postman both for exporting Postman Collections and mock serve.	10
SE1 Interpret available documentation when making use of a third-party API.	10
SE2 Manage API replies whilst displaying content in an interactive product.	10
Total Mark	100

Notes to Students:

- This assignment brief has been approved and released by the Internal Verifier through Classter.
- Assessment marks and feedback by the lecturer will be available online via Classter (http://mcast.classter.com) following release by the Internal Verifier
- Students submitting their assignment on Moodle/Turnitin will be requested to confirm online the following statements:

Student's declaration prior to handing-in of assignment

- I certify that the work submitted for this assignment is my own and that I have read and understood the respective Plagiarism Policy

Student's declaration on assessment special arrangements

- I certify that adequate support was given to me during the assignment through the Institute and/or the Inclusive Education Unit.
- I declare that I refused the special support offered by the Institute.

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Purpose and Aims

Application Programmable Interfaces (APIs) provide programmers with an abstraction of the implementation of complex systems. This allows users to make simple use of underlying complex systems, without having to know about the complexities of the system itself. By the end of the unit, learners will have a clear idea of what APIs are, why they exist and the different types of APIs available. They will also have knowledge on how to design, develop and document RESTful APIs, following best practices in terms of architectural and schema design, as well as taking into consideration the security aspect of an API.

Scenario

For this assignment, you will be going through the complete process of building an API for a system of your choice; starting from the planning phase, developing the API itself based on that planning, and testing the API when the development phase is over. You are to come up with an idea for a system that will have a public facing Web Application, an Administrator Back-End System, as well as a Mobile Application and/or Video Game. Based on this concept, you are to build an API that is capable to serve all these Front-End systems.

NB: You are encouraged to employ skills you gained through other units so far wherever you deem it to help improve your work for this assignment.

Learning Outcomes	
LO1	Differentiate the requirements of APIs and common security measures
LO2	Determine implementation requirements for an API following best practices
LO3	Develop an API following the implementation requirements and design criteria
LO4	Prepare mock servers and relevant documentation

Important Notes:

- **Private repositories will not be accepted for grading.**
- **Marks will be lost if documents are submitted in any other format than PDF.**
- **Late submissions, irrespective of how late, will be capped at a 50% mark.**

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Task 1 – Research Document

For this task, you need to come up with an idea for a system that would require an API to function to its highest potential. You are to come up with an idea for a system that that would benefit greatly from the creation of an API. You API needs to cover a vast variety of functions that would be useful for any external systems, including 3rd party systems, that would need to use this API.

It is suggested (but not mandatory) that the API created for this unit would be in line with the scope of one or more units you have completed or are currently working on. For example, you can create an API for the that would create a bridge between the PHP web application and the Ionic App created during the previous semester, or link it to a game your developed during the previous semester which could like to a website or other 3rd party services.

If you want to create an API that is not related to a previous project, the following are a few examples of APIs that could be created.

- Restaurant Booking System
- Concert Hall Event System
- Task Management System
- Video Game Community System
- Financial Organisation
- Budgeting and Expense Tracking
- Music Library
- Customer Relationship Management
- Video Game achievements
- Video Game statistics

In this task, you are to conduct some research and write up a planning document for your proposed API. In a document of not less than 1000 words, you are required to include the following.

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Part 1 – RESTful APIs (KU1)

Provide a brief description of what a RESTful API is. As part of this research, make sure to discuss the following:

- Core principles (Client-Server Architecture, Statelessness, Cacheability, etc)
- How making a request to an API works (request / response process)
- HTTP Methods (GET, POST, PUT, PATCH, DELETE)
- QueryString Parameters and Request Body

Based on this research, list a few examples of Endpoints you intend to implement in your API, including which HTTP methods they will be using and any QueryString Parameters and Request Body structures to be used for these Endpoints.

Part 2 – Authorisation (KU2)

Describe what OAuth 2.0 is, the benefits of implementing such an authorisation system, as well as how it is implemented. Make sure to go into some detail about the following:

- Scopes
- Access Tokens
- Client Id and Client Secret

Furthermore, make sure to provide an explanation of how you intend to implement this system within your own API.

Part 3 – Security Considerations (KU3)

Describe what the OWASP API Security Top 10 is. Making sure to use official sources, identify and explain the latest security threats listed within the top 10.

Based on this list, you are to select at least 2 out of this list and propose how you will mitigate these threats in your own API.

Submission: Research Document in PDF Format uploaded to a **Public GitHub Repository**

Deadline: 27th March 2026

Grading Criteria: *KU1, KU2, KU3*

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Task 2 – Project Development & Testing

For this task, you will be developing your API project. Before development is initiated, although not required, you should construct an ERD of the database that will be used by your API, as this will help you when you move on to construct the database which will be used by your API. This should include all the necessary information, including data types, constraints, and relationships.

Part 1 – API Back-End Development (AA1)

Your API must successfully connect to a database, and your classes are to be constructed in a way that your planned functionality is included. All the methods that interact with the database should be within the appropriate class, making sure that the correct access modifiers (public/private) are used.

Part 2 – Endpoints (SE2)

Construct all the necessary endpoints which use the classes and methods included in the API, giving access to all the functionality of the API to any external system that might need them. The responses of these endpoints must be properly formatted in JSON, using the correct HTTP response codes.

Part 3 – Consuming an API through cURL (AA2)

You are to build a small PHP web application that makes use of cURL to consume some of the endpoints within your API. Make sure to use some endpoints of each Request Type. The design of this application is not the focus for this unit but make sure that the results of these endpoints are displayed in an understandable manner.

Part 4 – Testing (AA4)

Postman will be used to test your API endpoints. Using Postman, you are to create a Workspace for all the requests related to your API. Inside this Workspace, you are to create multiple Collections, one for each type of resource. At least one API request to each endpoint is required in these collections. However, if an endpoint can handle requests with different parameters, that endpoint should have multiple requests saved in the relevant Collection.

Once all is done, you are to export all your Collections, and add them to the project's repository, inside a folder dedicated for your Postman Collections. Make sure that all Collections are included.

Important Notes

Make sure your code includes inline comments, explaining what each Class, Method, and Endpoint is used for. Make sure to add descriptive summaries to your Commits. Make sure your Commits are created at regular intervals; do not Commit all your work at the end of semester.

Submission: Project files, Database Export, and Postman Export added to the **same repository submitted for Task 1**

Deadline: 15th May 2025

Grading Criteria: *AA1, AA2, AA4, SE2*

Task 3 – Documentation & Evaluation

For your final task, you will be documenting your project and your development process.

Part 1 – Project Documentation (AA3)

Add a ReadMe file to your repository, making sure to cover all the important parts of your project. Remember that this documentation is meant to help you remember what you did and what everything does in your project, as well as aid prospective teammates in the future to understand the current structure of your project.

Part 2 – API Documentation (SE1)

Furthermore, you are to write a highly detailed documentation website using MkDocs. This documentation project needs to explain what each endpoint does and how it is used. Keep in mind that this is going to be used by other developers who need to consume your API to develop other projects.

Part 3 – Evaluation (KU4)

For this final part of this task, you will be critically evaluating your process for this project, from design, through development, testing, and exportation of results. In **not less than 800 words**, discuss how you went about developing your API, testing it via Postman, and how your API can be further improved in future development.

Submission: Evaluation Document in **PDF format**, **GitHub ReadMe**, and **MkDocs Documentation** website added to the **same GitHub repository submitted for Task 1 & 2.**

Deadline: 29th May 2025

Grading Criteria: *KU4, AA3, SE1*

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Minimum Evidence List

1	**T1:** 1 PDF Document via Public GitHub repository	☐
2	**T2:** Project Files via Public GitHub repository	☐
3	**T2:** Database Export via Public GitHub repository	☐
4	**T2:** Postman Export via Public GitHub repository	☐
5	**T3:** GitHub ReadMe file added to GitHub repository	☐
6	**T3:** MkDocs Documentation website via Public GitHub repository	☐
7	**T3:** Evaluation Document via Public GitHub repository	☐

Print this page and hand in with your assignment on final hand in date.